

VINANTH P

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EDUCATION

National Institute of Technology, Surat

B.Tech in Artificial Intelligence

Surat, Gujarat — CGPA: 8.07 / 10.0

Aug. 2024 – May 2028

Sri Chaitanya PU College

Central Board of Secondary Education

Bangalore, Karnataka — 12th: 93%

2022 – 2024

United Academy

Karnataka School Examination and Assessment Board

Hassan, Karnataka — 10th: 99.68%

2022

TECHNICAL SKILLS

Languages: Python, C, C++, HTML, CSS, JavaScript

ML Libraries: NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib, Seaborn, LangChain, LangGraph

Databases & Tools: SQL, Git, GitHub, Jupyter Notebook, Google Colab, VS Code, ChromaDB, Streamlit, MongoDB

Core Concepts: Supervised & Unsupervised Learning, Linear Algebra, Probability & Statistics, Data Structures & Algorithms, Operating Systems, Computer Networks, Retrieval-Augmented Generation (RAG)

PROJECTS

TCP Chat Room Application | C++, POSIX Sockets, Multithreading, CMake

- Built a multithreaded TCP chat server from scratch using POSIX socket APIs (socket, bind, listen, accept, send, recv) for reliable client-server communication
- Designed a custom application-layer messaging protocol with length-prefixed packets to correctly handle TCP byte-stream semantics and reliable message boundaries.
- Implemented concurrent client management using `std::thread`, `std::mutex`, and STL containers to safely broadcast messages across multiple connected clients.
- Structured the project with RAII principles, modular C++ design, and CMake-based cross-platform builds.

AI Video Assistant with RAG | Python, LangChain, Mistral AI, ChromaDB, Whisper, Streamlit, Sarvam AI

- Engineered a video/audio analysis tool capable of processing media files and YouTube URLs, utilizing local OpenAI Whisper for English transcription and Sarvam AI for Hindi/Hinglish.
- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain LCEL, ChromaDB, and HuggingFace Embeddings to enable users to interactively chat with meeting transcripts.
- Integrated Mistral AI to extract actionable meeting insights, and developed a Streamlit UI to export these comprehensive reports as PDF or TXT files.

Movie Recommender System | Python, Pandas, Scikit-learn, TMDB API, Streamlit

- Built a content-based movie recommendation engine using cosine similarity to suggest similar films from TMDB metadata.
- Engineered text features from movie titles, genres, cast, crew, and overview to improve recommendation quality.
- Integrated TMDB dataset preprocessing and feature extraction pipelines for scalable similarity computation.
- Deployed an interactive Streamlit interface for search, recommendations, and user-friendly movie discovery.

COMPETITIVE PROGRAMMING

- **LeetCode:** Solved 400+ problems across arrays, DP, graphs, trees, and binary search; consistent problem-solver at Easy-Medium level.
- **Codeforces:** Rated 1300+ (Pupil); regularly participates in Div. 2 and Div. 3 contests, strengthening algorithmic thinking and time-constrained problem solving.

SUMMARY

AI undergraduate with experience developing machine learning and Retrieval-Augmented Generation (RAG) applications using Python, PyTorch, LangChain, ChromaDB, and Streamlit. Strong understanding of data structures, algorithms, and core computer science fundamentals, complemented by competitive programming and hands-on software engineering projects.

